EB-JEPA — Reference Table (V1)

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June 2026

Our EB-JEPA runs on TUAB (v3.0.1, patient-disjoint, 276 eval recordings). *per-rec* = per-recording (16-window mean-pool, the clinical metric); *per-win* = per-window (the literature convention). ± = 3-seed standard deviation; other rows are single-seed. Companion to `explanation_augmented_corruption.tex` (which details the augmentation) and `PROVENANCE_TABLE.md` (the literature baselines).

#	**Encoder**	**Params**	**Energy (SSL objective)**	**Eval**	**per-rec BAcc / AUROC**	**per-win BAcc / AUROC**
baseline	Conv1D	0.4M	VICReg — *base aug, no corruption*	frozen	0.796 / 0.888	0.756 / –
1	Conv1D	0.4M	**VICReg + spectral 0.1**	frozen	0.836 / 0.887	0.765 / –
2	Conv1D	0.4M	**SIGReg**	frozen	0.825 / 0.913	**0.775 / 0.856**
3	Conv1D	0.4M	**VICReg**	frozen	**0.819 ±.004 / 0.900 ±.006**	0.770 / 0.848
4	Conv1D	0.4M	VICReg	FT	0.812 ±.004 / 0.908 ±.006	–
5	**Transformer** (multi-corpus)	3.65M	VICReg	frozen	0.798 / 0.872	0.738 / –
6	Conv1D	0.4M	**VICReg + spectral 0.1 +** $\Delta z/\Delta z^2$**/ACF/sACF**	frozen	0.823 ±.007 / 0.900 ±.010	0.761 ±.004 / 0.841 ±.004

Fixed for every corruption run (never swept): views v_1, v_2 made with Gaussian noise $\sigma = 0.1$, scale-jitter ±20%, channel-drop $p = 0.2$, plus exact EB-corruption — 20% timepoints masked + 20% $\pm 6\sigma$ outliers (≈40%/view). The **baseline** is the no-corruption reference (legacy contiguous time-mask).

Row 6 — stylized-facts energy (3-seed). Row 1's energy plus four view-consistency blocks between the two views' encoder feature maps, each an MSE between a path-statistic of v_1 and v_2, with explicit, fixed a-priori (not eval-swept) coefficients: increments Δz (0.10), realized variance $\mathrm{E}[\Delta z^2]$ (0.05), temporal autocorrelation over lags 1–8 (0.10), and autocorrelation of the power spectrum / FFT (0.05); the spectral term stays at its validated 0.10. All ≤ 0.1 so VICReg's variance/covariance anti-collapse terms remain dominant. *Result: squarely on the 0.82 plateau* — the added path-statistics regularise the representation but do not lift the 0.4M-conv ceiling (per-window is, if anything, ∼1 pp lower than row 3).